Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/05/2013
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Project Title:	Self-Structuring in Dusty Plasmas (NNX07AD22G)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	03/01/2007	**End Date:**	04/14/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
Contact Email:	ulf.e.israelsson@jpl.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Monitoring NASA Center changed to JPL from GRC (Ed., 6/2013) NOTE: Received funding continuation to 4/14/2013, per NSSC (8/2010) NOTE: Received extension to 2/28/11 (4/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX07AD22G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Fundamental physics research is carried out by performing experiments with dusty plasmas. Ground-based laboratory experiments are carried out to prepare for future microgravity flight experiments, including the planned European PK-4 instrument. Data were analyzed from the predecessor project PK-3 Plus on the International Space Station (ISS). (The acronym PK refers to "Plasma Crystal.")
Dusty plasma consists of particles of solid matter suspended in ionized gas. Micron-size particles acquire a large negative charge of typically 10,000 elementary charges. Due to inter-particle repulsion, they arrange themselves in structure with solid or liquid-like properties, similar to a colloidal crystal. With particles of this size, gravity is a dominant force, causing particles to sediment to a thin layer at the bottom of the plasma.

Under microgravity conditions, particles fill a 3D volume and the suspension self-organizes. We study:

• Structural features. These include the boundaries of the particle suspension, a void at the plasma's center, and crystalline or liquid-like microstructure in the arrangements of particles.

• Dynamical behavior exhibited by the dusty plasma. These include waves, instabilities, transport, and random motion associated with a non-equilibrium system. Dynamical behavior is studied at an atomistic level, using video microscopy to observe the motion of microspheres which serve as an analog for atoms in normal molecular matter.

Experimental methods include direct imaging of particle suspensions using video microscopy and particle tracking, and manipulation of particles using the radiation-pressure force of powerful cw lasers.

Ground-based laboratory experiments and numerical simulations accompany the analysis of data returned from space. This ground-based work serves the dual purposes of developing scientific methods to be used in future flight experiments as well as providing an immediate scientific return. This ground-based work is mainly performed in the PI's laboratory at The University of Iowa, but it can also include analysis of data from parabolic-flight experiments conducted by Europeans.

The PK-4 mission is an ESA project, under the direction of ESA. The PI collaborates with the ESA scientist who is responsible for the mission. The PI is a member of ESA's Facility Science Team for the PK-4 mission.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

During this period, Professor Goree's group carried out ground-based research to support future flight experiments. This research included laboratory experiments, numerical simulations, and collaboration on parabolic flight experiments.
Laboratory experiments:

Dust density wave: The dust density wave, also known as the dust acoustic wave, is a low-speed compression wave. It is expected to be observed in PK-4, so our laboratory experiments help prepare for planning PK-4 experiments by identifying experimental methods and data analysis methods. We reported a laboratory experiment with the dust-density wave in which we characterized nonlinear synchronization. In synchronization, a natural oscillation (like the dust density wave) that tends to occur at frequency f0 will shift its frequency if there is a periodic external disturbance at a different driving frequency. We carried out laboratory experiments and found several synchronized states, including the simple 1:1 state where the oscillation frequency shifts to match the drive. We discovered features for this synchronization that have not been previously reported, including a threshold amplitude for the drive and a branching feature in the Arnold tongue diagram for the synchronization. This work was published in Physical Review E. We expect that the PK4 instrument may allow a similar synchronization experiment under microgravity conditions. Microgravity conditions will allow operating with a longer 3D dust cloud than is possible under 1g conditions, so that synchronization over a large number of wavelengths can be studied.

Shear flows and viscous heating: In PK4 it will be possible to drive a shear flow in the 3D dust cloud by applying the radiation pressure force using a manipulation laser. To develop physics concepts that can be studied experimentally by using this manipulation laser, we carried out laboratory experiments to observe laser-driven shear flows in a 2D suspension of dust particles. We found that we were able to characterize the local temperature increase due to viscous heating, in a region of high shear flow. Importantly, this was the first experimental detection of local heating in a velocity gradient. It has been known for over a century that viscous heating raises the temperature in an entire flow, for example in a pipe, but the effects of thermal conduction usually make it difficult to measure a local temperature peak in a shear flow in an ordinary liquid like water. By using a liquid-like dusty plasma, we were able to observe this long-predicted phenomenon for the first time, and the result was published in Physical Review Letters. A follow-up paper was published in Physical Review E.

Numerical simulations

3D Code: Planning for PK4, we developed a new molecular-dynamics code that is useful for simulating a 3D cloud of particles. This work was done with assistance from a DLR-funded German colleague. In our first use of the simulation, we computed particle trajectories under conditions identical to those reported the Russian group Fortov et al. for a breadboard PK4 instrument. We computed the diffusion coefficient and viscosity coefficient. The results for these coefficients are expected to be suitable for direct comparison to the future space-based experiments. We also examined the feasibility of tracking particles, using the PK4 cameras, with sufficient accuracy and time duration for performing similar calculations with PK4 experimental data.

2D Code: Using a previous 2D codes, we carried out simulations to characterize viscoelastic properties of 2D clouds of dust particles in a plasma. Three papers were published in Physical Review E, one of them with collaborators in Hungary. The physics concepts of viscoelasticity in a dusty plasma are different from those in colloidal suspensions, for example, due to the lower level of friction applied by the rarefied gas in a dusty plasma as compared to the solvent in a colloidal suspension. We expect that it may be possible to characterize under microgravity conditions some of the same measures of viscoelasticity that we studied numerically. These measures include the cutoff wave number, Maxwell relaxation time, frequency-dependent shear viscosity, and longitudinal viscosity.

Microgravity parabolic flight analysis: In collaboration with DLR-funded German researchers, microgravity data was analyzed for parabolic flights that were conducted by the Germans. Results were reported in a paper published in Physics of Plasmas. In this paper, we reported the discovery of an unexpected feature in the cloud of dust particles:

	chains of particles that were aligned parallel to the ambient ion flow. This was unexpected because the ion flow velocity was subsonic. It was previously known that chains form in supersonic and nearly supersonic flows due to forces applied by the ion wake downstream of a dust particle, but in this experiment we found such chains at very low Mach numbers of order 0.1. It was found that these chains most often appear in regions of low number density of dust. It was hypothesized that a previously unidentified attractive force might be responsible for this chain formation. Our paper in Physics of Plasmas immediately attracted theoretical attention, and a scientist at MIT published a computer simulation attempting to identify the source of the attractive force. We expect to look for evidence of this effect in the forthcoming PK4 experiment.
Bibliography Type:	Description: (Last Updated: 01/23/2025)
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